

D365 Finance and Operations Testing Strategy and Methodology.

Streamlining Business Processes with Thorough Testing

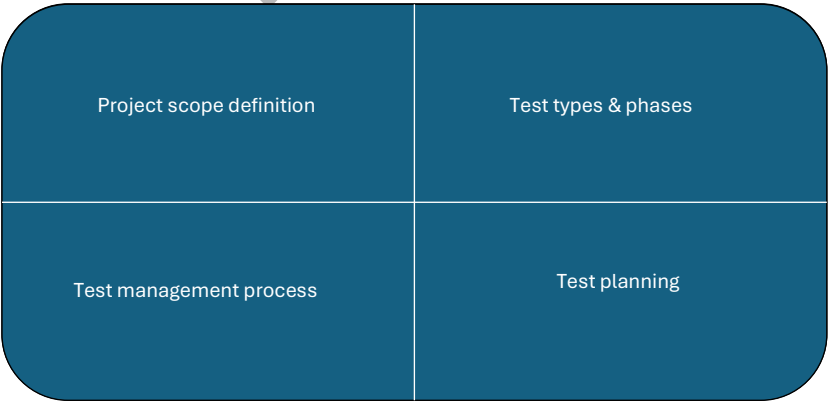
Introduction

In today's fast-paced business environment, implementing Microsoft Dynamics 365 Finance and Operations (D365 F&O) is crucial for streamlined processes and better decision-making. A well-structured testing strategy is essential to ensure the system functions optimally and meets business needs. This article examines the testing strategy and methodology for D365 F&O implementation.

Why do we test?

• Ensure business can operate safely and efficiently in Production • Understand the behavior of the solution • Validate design • Drive feedback • Ensure stability and reliability • Validate data quality • Ensure solution works	• Improve solution quality • Improve user experience • Validate performance • Improve code design • Find bugs • Validate requirements • Ensure stability and reliability • Validate security and compliance
--	---

Test strategy components



The Significance of Testing in D365 F&O Implementation

Testing is crucial in the D365 F&O implementation lifecycle. It identifies and mitigates issues, ensuring the system works as expected. Effective testing aligns all modules with business processes, reducing post-implementation risks and improving user adoption.

Testing Strategy for D365 F&O Implementation

A comprehensive testing strategy encompasses several critical components that guarantee thorough validation of the D365 F&O system.

1. Defining Testing Objectives

The first step in creating a testing strategy is to set clear objectives that align with the D365 F&O implementation goals. Objectives should include validating system functionality, ensuring data integrity, verifying integration with other systems, and confirming performance meets business needs.

2. Key Stakeholders Identification

Engaging key stakeholders early is vital for successful testing. This includes project managers, business analysts, developers, and end-users. Their involvement ensures all requirements are captured and feedback is considered throughout the process.

3. Creating a Test Plan

A test plan details the scope, approach, resources, and schedule for testing. It should include unit, integration, system, user acceptance (UAT), and performance testing phases.

4. Setting Up Test Environments

Dedicated test environments that replicate the production setup are crucial for accurate testing. They should include all essential hardware, software, and network configurations to simulate real-world scenarios.

5. Defining Test Cases

Detailed test cases based on business requirements and use cases are crucial for thorough testing. They should cover all scenarios, including positive and negative paths, to ensure the system handles different situations effectively.

Project scope mapped to test scope

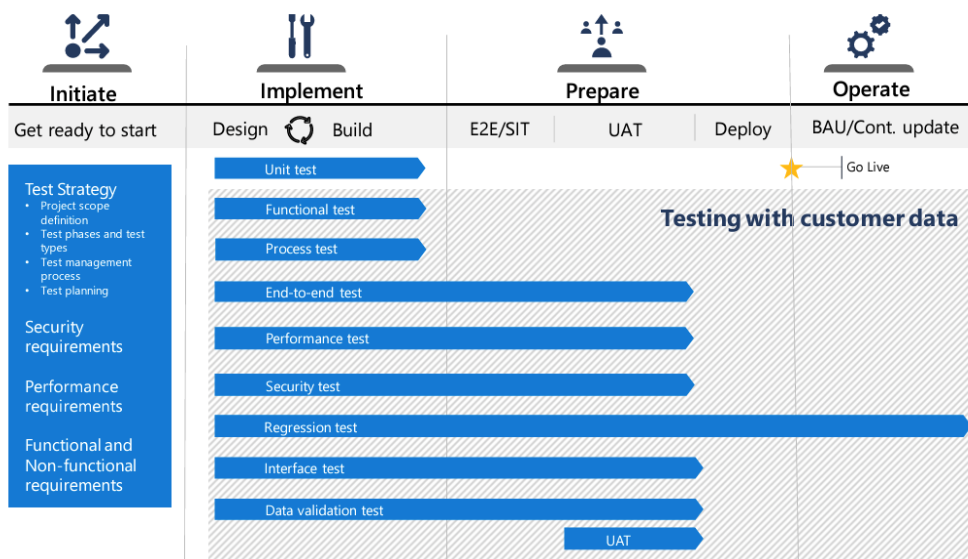
- A clearly defined project scope helps to define the test scope.
- End-to-end business processes provide context for requirements and test suitability.
- Many aspects of project scope help determine the required test coverage.

	Business processes
	Data – Integrations, Migration, Reporting
	Batch processes
	Geographical scope
	Security (e.g. Project roles)
	Complex and custom designs
	Non-functional : performance, infra, usability etc
	Impact of process changes, Regulatory reqs
	Business goals

Attributes of tests and test phases

Objectives of this test	Intended coverage and source for test case
When this test type/phase should be used	Who owns this test type/phase
Preparation needed to meet the entry criteria	Project roles and resources to prepare, perform and report on the test type/phase
Entry criteria for this test	Tools that can/should be used for this test
How testing should be performed/managed	Planning considerations
Exit criteria for this test	Relationship with other testing types/phases

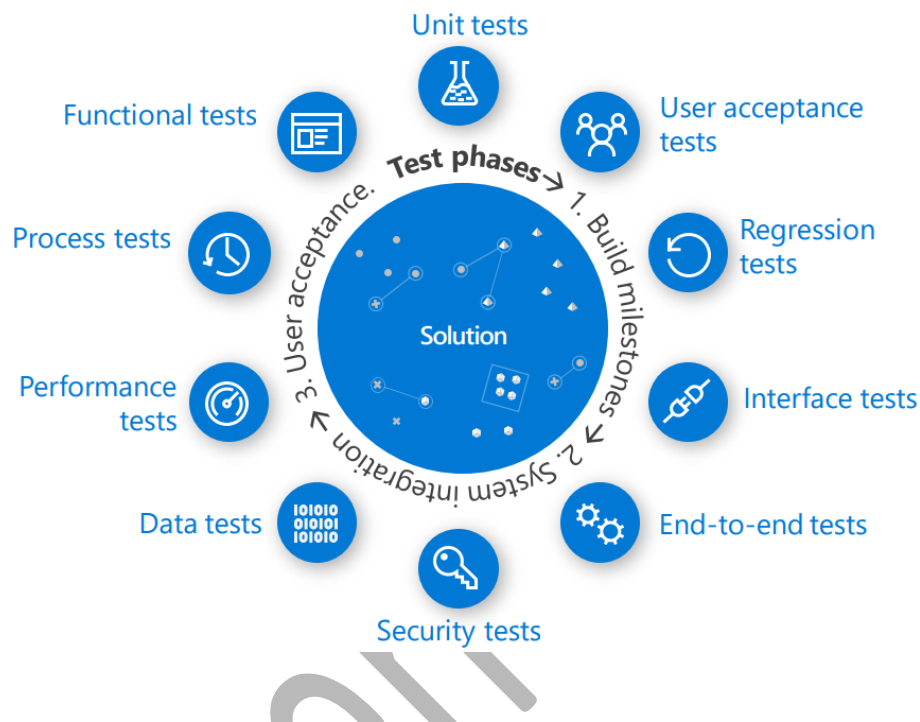
Testing throughout the solution lifecycle



Testing Methodology for D365 F&O Implementation

The testing methodology describes the processes and techniques used to perform testing activities. A comprehensive method for D365 F&O implementation includes the following phases:

Test types and test phases



Test type	What are we trying to answer
Mock go-live	Does my cutover process work and am I confident the production go-live will go smoothly
Smoke tests	Very basic functions work as expected (e.g. login or running reports etc)
Data acceptance tests	Did my data migrate correctly and can it be inquired upon, transacted upon and reported upon
Interface tests	Can my interfaces be consumed as intended.
Environment testing	Validating that a new environment is fit for purpose
Reporting/BI testing	Specific test type to validate reporting/BI including from external systems such as data warehouses, data lakes etc
Device testing	Specific testing to validate shop-floor and warehouse devices such as scanners, POS, RFID readers etc Can include mobility device testing such as phones, tablets etc..
Network/Infrastructure testing	Validation of any relevant underlying networking, fire wall configuration, Wifi, printers etc

Tools for test and data setup

Tool	Purpose
Perf SDK	Single and multiuser performance testing
Data Expansion Tool	Load testing, created high volume data for load testing
Data Entities	Data management framework for data
Data Task Automation	Validation of task outcomes
Regression Suite Automation Tool (RSAT)	Automated regression testing based on task recordings without writing code
Acceptance Test Library (ATL)	Mostly for data setup and validation in integration and component tests
Azure DevOps	Manage requirements, tests, defects, etc.
SysTest Framework	Unit tests for regression testing, in Visual Studio
Third-party tools	Postman, JMeter, Leapwork, etc.

1. Unit Testing

Unit testing verifies individual components of the D365 F&O system to ensure they work correctly in isolation. Developers use automated tools to streamline this process and catch issues early.

Purpose: Validate the functionality of individual components or modules.

- Tools: Visual Studio, X++ Test Framework
- Scope: Isolated testing of specific functions or methods
- Approach: Developers create and run unit tests to catch issues early during development.

2. Integration Testing

Integration testing checks interactions between modules and external systems, ensuring accurate data flow. Developers and testers work together to fix integration issues.

Purpose: Ensure that different modules or systems work together as expected.

- Tools: Azure DevOps, Postman, Data Management Framework (DMF)
- Scope: Test interactions between integrated components
- Approach: Conduct testing to validate data flow and interactions between modules.

3. Functional Testing

Functional testing validates the D365 F&O system as a whole. It ensures all modules and functions work correctly in an integrated environment. Testers run end-to-end scenarios to confirm the system meets business requirements.

Purpose: Confirm that the system behaves as expected under various scenarios.

- Tools: Task Recorder, RSAT (Regression Suite Automation Tool)
- Scope: End-to-end business processes
- Approach: Use predefined test cases to verify that the system meets functional requirements.

4. User Acceptance Testing (UAT)

User Acceptance Testing (UAT) is where end-users validate the system with real-world scenarios. It ensures the system meets user expectations and performs well in daily operations. Engaging users helps identify usability issues and gather feedback for improvements.

Purpose: Validate the system from the end-user's perspective.

- Tools: Azure DevOps, Microsoft Teams
- Scope: Real-world user scenarios
- Approach: Involve end-users to test the system and provide feedback on usability and functionality.

5. Performance Testing

Performance testing measures D365 F&O's responsiveness, stability, and scalability under different conditions. It ensures the system handles peak loads efficiently. This involves simulating user loads and tracking response times, throughput, and resource use.

Purpose: Assess system performance under load conditions.

- Tools: Azure Load Testing, JMeter
- Scope: System's response times, throughput, and resource utilization
- Approach: Simulate real-world usage scenarios to measure performance and identify bottlenecks.

6. Regression Testing

Regression testing ensures new changes don't negatively impact existing functions. It involves re-running past tests to confirm the system works correctly after modifications. Automated tools help speed up this process.

7. Security Testing

Security testing determines if the D365 F&O system is secured against potential threats and vulnerabilities. This phase involves evaluating access controls, data encryption, and compliance with industry standards. Implementing strong security measures is necessary for protecting sensitive business data.

Purpose: Identify vulnerabilities and ensure data protection.

- Tools: Azure Security Center, Penetration Testing Tools
- Scope: Access controls, data encryption, and compliance
- Approach: Conduct vulnerability assessments and penetration tests to safeguard data.

Comprehensive Matrix

Testing Phase	Tools	Scope	Approach
Unit Testing	Visual Studio, X++ Test Framework	Isolated components	Developers create and run unit tests
Integration Testing	Azure DevOps, Postman, DMF	Component interactions	Validate data flow and interactions
Functional Testing	Task Recorder, RSAT	End-to-end processes	Use predefined test cases
Performance Testing	Azure Load Testing, JMeter	System performance	Simulate real-world scenarios
Security Testing	Azure Security Center, Penetration Testing Tools	Access controls, data encryption	Vulnerability assessments and penetration tests
User Acceptance Testing	Azure DevOps, Microsoft Teams	Real-world user scenarios	End-users validate system usability and functionality

Best Practices for D365 F&O Testing

Implementing established practices can improve the testing strategy and methodology for D365 F&O. Important practices include:

- **Test Plan:** Overall, the test plan within the project plan includes all the test phases and test types.
- **Effort and Resources:** Include effort and resources for each test phase, including the estimated test iterations.
- **Dependencies:** Beyond the apparent functional readiness, consider related dependencies.
- **Environmental strategy:** Proper granularity in environment strategy provides higher flexibility and speed/quality of implementation. Excessive granularity in environment strategy can be challenging to maintain and wastes resources.
- **ALM Context:** Code management, software packaging, troubleshooting, and bug fixing.
- **Test output:** Consider the retest/iteration outcomes based on results in exit criteria.
- **Early Tester Involvement:** Engaging testers early helps spot issues sooner and ensures they understand business requirements well.
- **Automation of Testing:** Leveraging automated testing tools can expedite testing processes, increase coverage, and reduce manual effort. Automation is particularly beneficial for regression and performance testing.
- **Continuous Testing:** Adopting a constant testing approach allows for ongoing system validation throughout the implementation lifecycle. Continuous testing helps identify and address issues promptly.
- **Comprehensive Documentation:** Maintaining detailed documentation of test cases, test results, and issues encountered ensures transparency and facilitates future reference.
- **Consistent Communication:** Developing consistent communication practices among stakeholders, testers, and developers promotes collaboration and ensures alignment with testing objectives and progress.
- **Update and Version:** Adopt the "One Version" policy and plan for 2 nevertheless two major updates per year.

Conclusion

A well-defined testing strategy and methodology are essential for the successful deployment of D365 Finance and Operations. Adopting a structured approach to testing enables organizations to confirm that the system aligns with business requirements, operates efficiently, and delivers the intended benefits. The application of best practices further optimizes the testing process, contributing to a seamless and successful implementation of D365 F&O.

Hope this information helps!!!

Regards,

Hiten Parekh